

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

#1

```
library(here); library(tidyverse); library(agricolae); library(ggplot2); library(lubridate)
```

```
## here() starts at /home/guest/R/EDE_Fall2025/EDE_Fall2025
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats    1.0.0      v stringr    1.5.1
```

```
## v ggplot2    4.0.0      v tibble     3.2.1
```

```
## v lubridate  1.9.3      v tidyr      1.3.1
```

```
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()    masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```

here()

## [1] "/home/guest/R/EDE_Fall2025/EDE_Fall2025"

NTL_LTER_raw = read.csv(
  file = here('Data', 'Raw', 'NTL-LTER_Lake_ChemistryPhysics_Raw.csv'),
  stringsAsFactors = TRUE
)
NTL_LTER_raw$sampldate = as.Date(NTL_LTER_raw$sampldate, format = "%m/%d/%y")

#2
mytheme = theme_bw()+
  theme(plot.title = element_text(hjust = 0.5),
        plot.subtitle = element_text(hjust = 0.5),
        legend.position = "right"
  )
set_theme(mytheme)

```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question:

Answer:

H_0 : Mean July temperature is the same at all depths across lakes.

H_a : Mean July temperature differs with depth and/or across lakes.

4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: `lakename`, `year4`, `daynum`, `depth`, `temperature_C`
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```

#4
NTL_LTER = NTL_LTER_raw %>%
  filter(month(sampldate) == 7) %>%
  select('lakename', 'year4', 'daynum', 'depth', 'temperature_C') %>%
  drop_na()

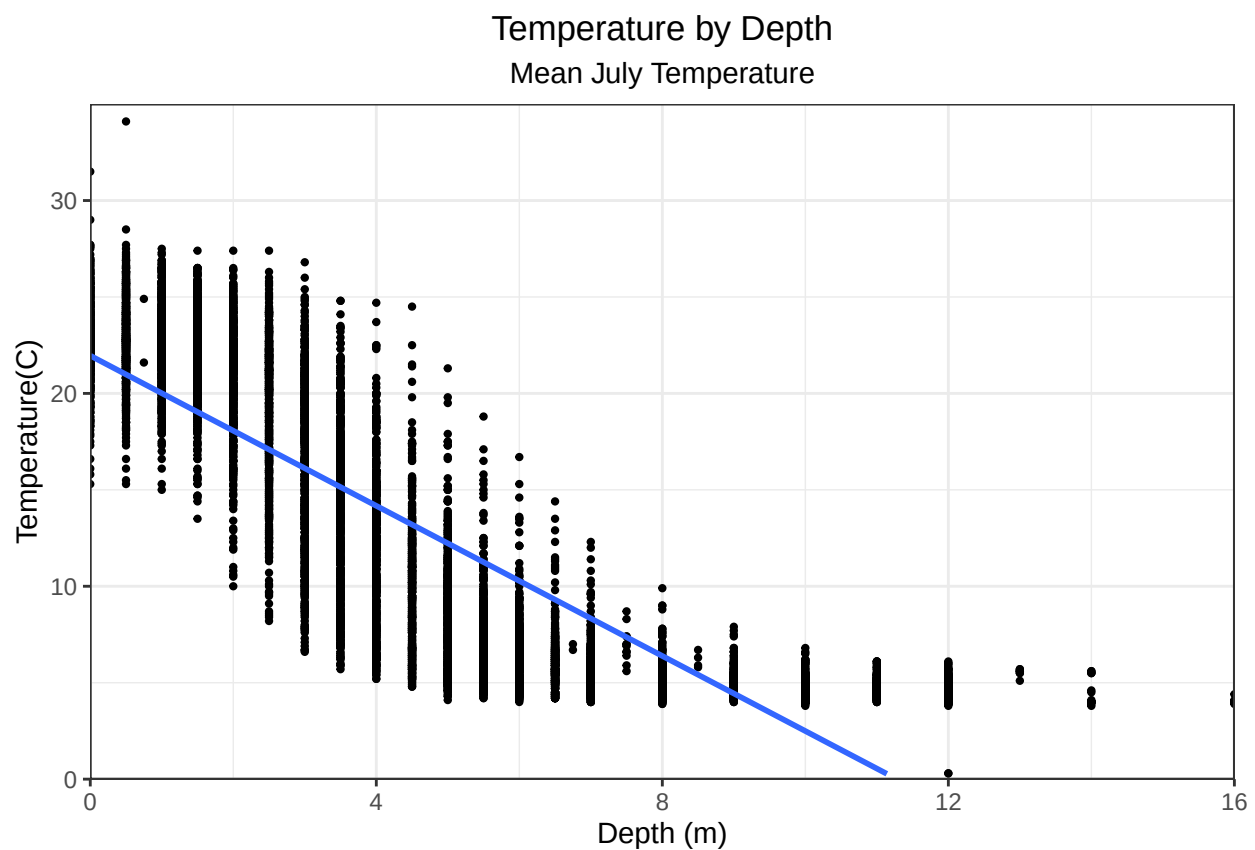
#5
ggplot(NTL_LTER, aes(x = depth, y = temperature_C)) +
  geom_point(size = 0.8) +
  geom_smooth(method = "lm", se = FALSE) +
  scale_y_continuous(expand = c(0, 0), limits = c(0, 35)) +

```

```
scale_x_continuous(expand = c(0,0))+
labs(
  title = "Temperature by Depth",
  subtitle = "Mean July Temperature",
  x = "Depth (m)",
  y = "Temperature(C)",
)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: The blue regression line suggests a strong negative trend overall. The distribution of points shows some curvature or nonlinearity — temperatures decrease rapidly in the upper few meters, then level off at greater depths.

7. Perform a linear regression to test the relationship and display the results.

```

#7
LR = lm(data = NTL_LTER, temperature_C ~ depth)
summary(LR)

##
## Call:
## lm(formula = temperature_C ~ depth, data = NTL_LTER)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  21.95597    0.06792   323.3  <2e-16 ***
## depth       -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF, p-value: < 2.2e-16

```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: Depth explains about 73.9% of the variability ($R^2 = 0.7387$). The result is based on 9,726 degrees of freedom, showing a large sample size and reliable estimates. The relationship is highly significant because ($p < 2.2 \times 10^{-16}$). For every 1-meter increase in depth, the mean July temperature decreases by approximately 1.95°C.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

```

#9
GLM = lm(data = NTL_LTER,
          temperature_C ~ year4 + daynum + depth)
step(GLM)

```

```
## Start: AIC=26065.53
## temperature_C ~ year4 + daynum + depth
##
##           Df Sum of Sq    RSS   AIC
## <none>                 141687 26066
## - year4    1         101 141788 26070
## - daynum   1         1237 142924 26148
## - depth    1      404475 546161 39189

##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = NTL_LTER)
##
## Coefficients:
## (Intercept)      year4      daynum      depth
##      -8.57556      0.01134      0.03978     -1.94644

#10
summary(GLM)
```

```
##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = NTL_LTER)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994  0.32044
## year4        0.011345   0.004299   2.639  0.00833 **
## daynum       0.039780   0.004317   9.215 < 2e-16 ***
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF, p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The AIC model selection procedure identified year, day number, and depth as the final set of explanatory variables. This model explains approximately 74.1% of the observed variance in temperature ($R^2 = 0.7412$). It is slightly improved as R^2 increased from 0.7387 to 0.7412.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality)

or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

#12

```
ANOVA = aov(data = NTL_LTER, temperature_C ~ lakename)
summary(ANOVA)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8  21642   2705.2     50 <2e-16 ***
## Residuals    9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ANOVA2 = lm(data = NTL_LTER, temperature_C ~ lakename)
summary(ANOVA2)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = NTL_LTER)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769  -6.614  -2.679   7.684  23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      17.6664     0.6501  27.174 < 2e-16 ***
## lakenameCrampton Lake      -2.3145     0.7699  -3.006 0.002653 **
## lakenameEast Long Lake     -7.3987     0.6918 -10.695 < 2e-16 ***
## lakenameHummingbird Lake  -6.8931     0.9429  -7.311 2.87e-13 ***
## lakenamePaul Lake        -3.8522     0.6656  -5.788 7.36e-09 ***
## lakenamePeter Lake       -4.3501     0.6645  -6.547 6.17e-11 ***
## lakenameTuesday Lake     -6.5972     0.6769  -9.746 < 2e-16 ***
## lakenameWard Lake        -3.2078     0.9429  -3.402 0.000672 ***
## lakenameWest Long Lake   -6.0878     0.6895  -8.829 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

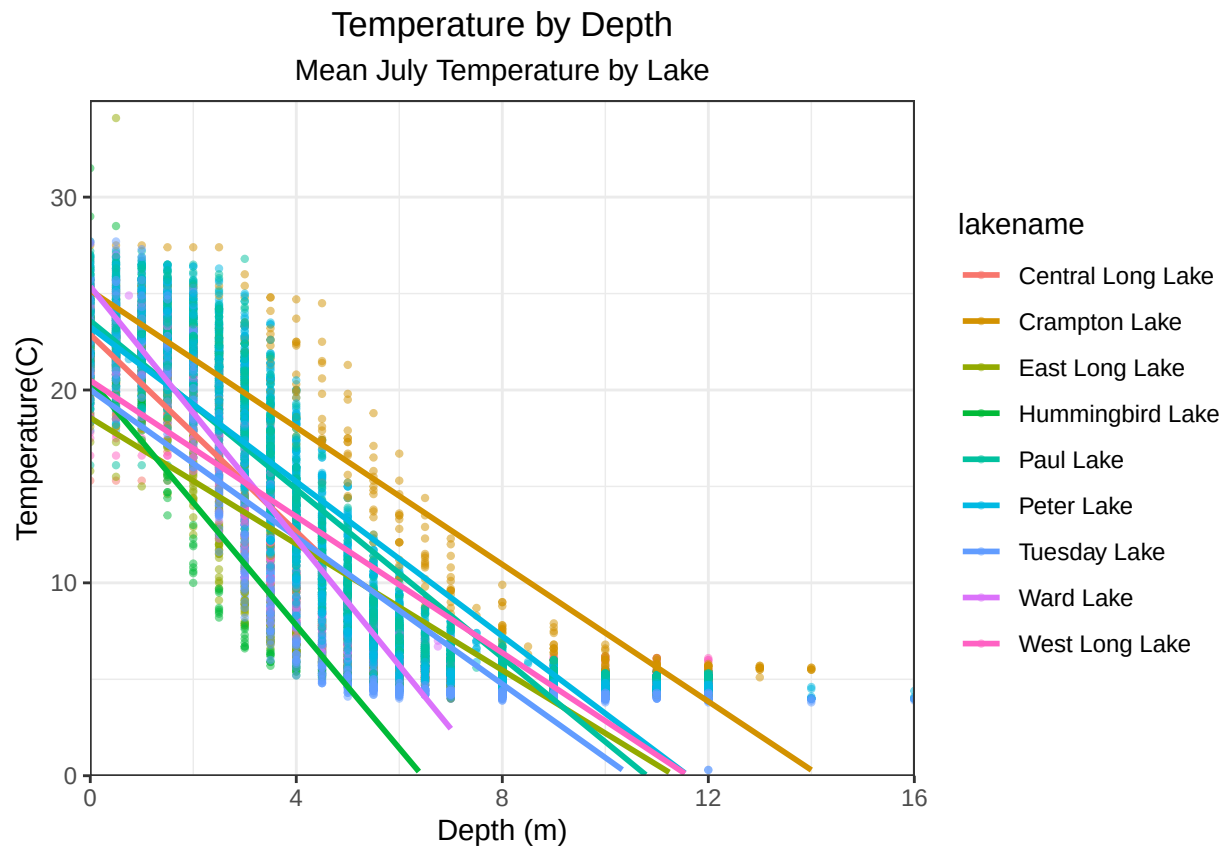
Answer: Mean lake temperature differs significantly among the nine lakes included in the analysis with $F(8,9719)=50$, $p < 2.2 \times 10^{-16}$. The extremely small p-value indicates that these differences are highly unlikely to have occurred by chance.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14.
ggplot(NTL_LTER, aes(x = depth, y = temperature_C, color = lakename))+
  geom_point(size = 0.8, alpha = 0.5)+
  geom_smooth(method = "lm", se = FALSE)+
  scale_y_continuous(expand = c(0,0), limits = c(0, 35)) +
  scale_x_continuous(expand = c(0,0))+
  labs(
    title = "Temperature by Depth",
    subtitle = "Mean July Temperature by Lake",
    x = "Depth (m)",
    y = "Temperature(C)",
  )
)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 73 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



15. Use the Tukey's HSD test to determine which lakes have different means.

```
#15
TukeyHSD(ANOVA)
```

```
## Tukey multiple comparisons of means
```

```
##      95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = NTL_LTER)
##
## $lakename
##              diff          lwr          upr          p adj
## Crampton Lake-Central Long Lake -2.3145195 -4.7031913  0.0741524 0.0661566
## East Long Lake-Central Long Lake -7.3987410 -9.5449411 -5.2525408 0.0000000
## Hummingbird Lake-Central Long Lake -6.8931304 -9.8184178 -3.9678430 0.0000000
## Paul Lake-Central Long Lake -3.8521506 -5.9170942 -1.7872070 0.0000003
## Peter Lake-Central Long Lake -4.3501458 -6.4115874 -2.2887042 0.0000000
## Tuesday Lake-Central Long Lake -6.5971805 -8.6971605 -4.4972005 0.0000000
## Ward Lake-Central Long Lake -3.2077856 -6.1330730 -0.2824982 0.0193405
## West Long Lake-Central Long Lake -6.0877513 -8.2268550 -3.9486475 0.0000000
## East Long Lake-Crampton Lake -5.0842215 -6.5591700 -3.6092730 0.0000000
## Hummingbird Lake-Crampton Lake -4.5786109 -7.0538088 -2.1034131 0.0000004
## Paul Lake-Crampton Lake -1.5376312 -2.8916215 -0.1836408 0.0127491
## Peter Lake-Crampton Lake -2.0356263 -3.3842699 -0.6869828 0.0000999
## Tuesday Lake-Crampton Lake -4.2826611 -5.6895065 -2.8758157 0.0000000
## Ward Lake-Crampton Lake -0.8932661 -3.3684639  1.5819317 0.9714459
## West Long Lake-Crampton Lake -3.7732318 -5.2378351 -2.3086285 0.0000000
## Hummingbird Lake-East Long Lake  0.5056106 -1.7364925  2.7477137 0.9988050
## Paul Lake-East Long Lake  3.5465903  2.6900206  4.4031601 0.0000000
## Peter Lake-East Long Lake  3.0485952  2.2005025  3.8966879 0.0000000
## Tuesday Lake-East Long Lake  0.8015604 -0.1363286  1.7394495 0.1657485
## Ward Lake-East Long Lake  4.1909554  1.9488523  6.4330585 0.0000002
## West Long Lake-East Long Lake  1.3109897  0.2885003  2.3334791 0.0022805
## Paul Lake-Hummingbird Lake  3.0409798  0.8765299  5.2054296 0.0004495
## Peter Lake-Hummingbird Lake  2.5429846  0.3818755  4.7040937 0.0080666
## Tuesday Lake-Hummingbird Lake  0.2959499 -1.9019508  2.4938505 0.9999752
## Ward Lake-Hummingbird Lake  3.6853448  0.6889874  6.6817022 0.0043297
## West Long Lake-Hummingbird Lake  0.8053791 -1.4299320  3.0406903 0.9717297
## Peter Lake-Paul Lake -0.4979952 -1.1120620  0.1160717 0.2241586
## Tuesday Lake-Paul Lake -2.7450299 -3.4781416 -2.0119182 0.0000000
## Ward Lake-Paul Lake  0.6443651 -1.5200848  2.8088149 0.9916978
## West Long Lake-Paul Lake -2.2356007 -3.0742314 -1.3969699 0.0000000
## Tuesday Lake-Peter Lake -2.2470347 -2.9702236 -1.5238458 0.0000000
## Ward Lake-Peter Lake  1.1423602 -1.0187489  3.3034693 0.7827037
## West Long Lake-Peter Lake -1.7376055 -2.5675759 -0.9076350 0.0000000
## Ward Lake-Tuesday Lake  3.3893950  1.1914943  5.5872956 0.0000609
## West Long Lake-Tuesday Lake  0.5094292 -0.4121051  1.4309636 0.7374387
## West Long Lake-Ward Lake -2.8799657 -5.1152769 -0.6446546 0.0021080
```

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Peter Lake, Paul Lake ($p = 0.224$), and Ward Lake ($p = 0.783$) have statistically indistinguishable mean July temperatures ($p > 0.05$). No single lake stands out as having a mean temperature that is significantly different from all others

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: T-test.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
NTL_LTER = NTL_LTER %>%  
  filter(lakename == "Crampton Lake" | lakename == "Ward Lake")  
t.test(NTL_LTER$temperature_C ~ NTL_LTER$lakename)
```

```
##  
## Welch Two Sample t-test  
##  
## data: NTL_LTER$temperature_C by NTL_LTER$lakename  
## t = 1.1181, df = 200.37, p-value = 0.2649  
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0  
## 95 percent confidence interval:  
## -0.6821129 2.4686451  
## sample estimates:  
## mean in group Crampton Lake      mean in group Ward Lake  
##           15.35189              14.45862
```

Answer: The test does not find a statistically significant difference ($p = 0.2649$) between the mean temperatures of two lakes. Statistically speaking, we don't have enough evidence to say they are different, so we treat them as equal. Results from part 16 also suggest that the two lakes have similar mean July temperatures ($p = 0.9714459$).